What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Protein Synthesis Inhibitor–1 of 2: Mechanisms of Action

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Adapted from: Protein Synthesis Inhibitors

Listen to this Brick
0:00 / 20:37 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of protein synthesis inhibitors.
2. Discuss key pharmacokinetic parameters that are considered when using the protein synthesis inhibitors.
3. Recognize th[illegible]d with protein synthesis inhibitors.
4. Identify the common drug interactions associated with protein synthesis inhibitors.

CASE CONNECTION

KH tells you, "I can deal with a little bit of acne," she says, with an embarrassed look toward her mother. "But I've reached the point where I don't really like to go out because of my face. I'm a senior, and I don't want to look like this at graduation." KH washes her face daily with ordinary soap. Her face has severe pustular acne. You decide KH would benefit from treatment with oral tetracycline. You explain to her that the tetracycline is an antibiotic that will fight the inflammation and reduce the bacteria that are causing most of the trouble.

What adverse effects will you discuss with KH? Consider your answer as you read, and we'll revisit KH at the end of the brick.

GO TO CONCLUSION ↓

What Are Protein Synthesis Inhibitors? (LO#1)

Antibiotics work by halting essential cellular processes to kill bacteria. Broken down into its roots, the word antibiotic is interpreted as "opposing life." This trait can be beneficial against such living things as pathogenic bacteria, which endanger us. Like our own cells, bacteria synthesize protein using ribosomes, which are located in [illegible]ibosomes. Additionally, ribosomes comprise two subunits, one small and one large. In prokaryotes, the small subunit is 30S, while the large is 50S. Ribosomes also contain three important sites for peptide synthesis. These are A = Acceptor, P = Peptidyl, and E = Elongation sites, where individual amino acids are loaded and bonded.

Protein synthesis inhibitor (PSI) antibiotics can interact with the A, P, and E sites or bind to other parts of the ribosomal subunits to inhibit bacterial growth or kill the bacteria (Figure 1). PSIs can be divided into two classes based on which subunit they target, so there are 30S and 50S ribosome-inhibiting antibiotics.

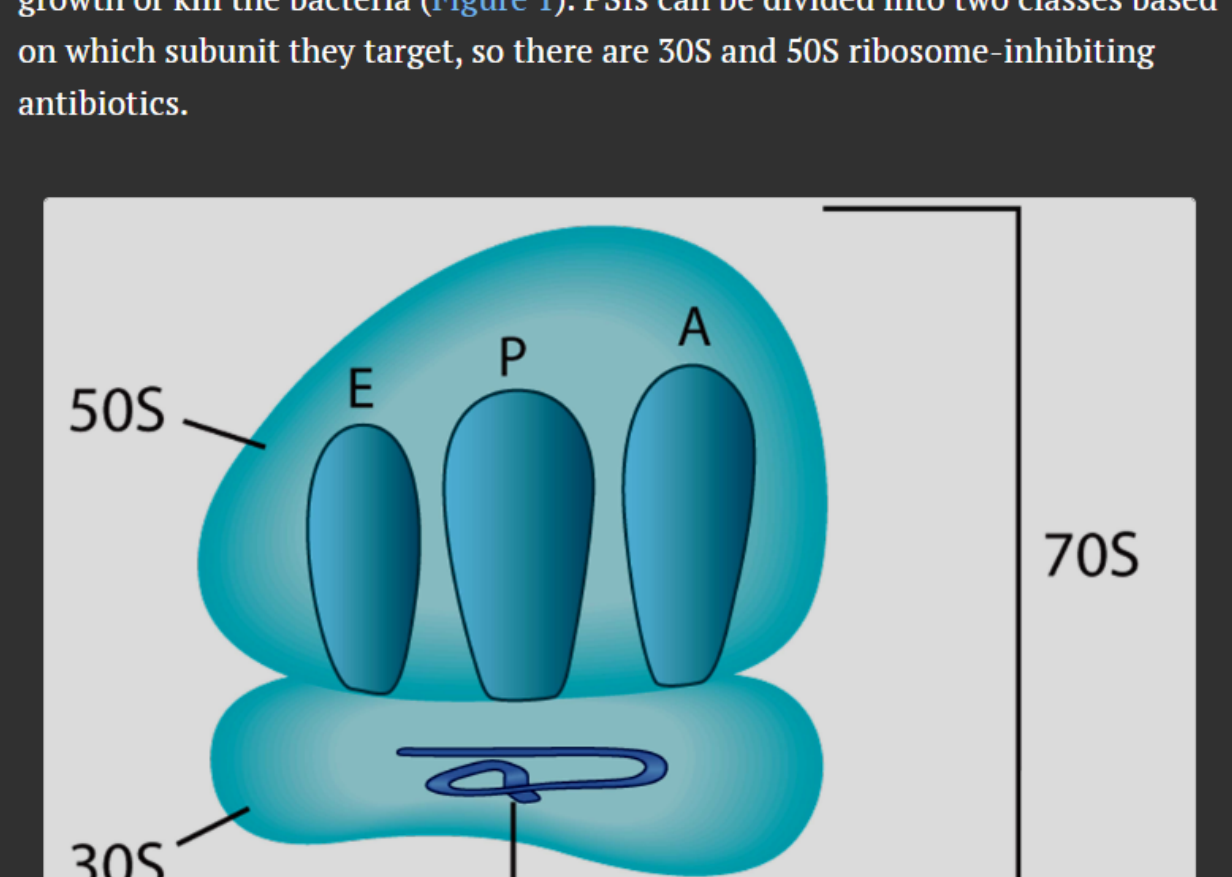


Figure 1 Bacterial r[illegible], E=Elongation site

What are the types of 30S Protein Synthesis Inhibitors?

Aminoglycosides and tetracyclines are two important members of the 30S PSIs. Structurally, aminoglycosides are composed of two or more amino sugars bonded to an aminocyclitol nucleus, and the various aminoglycosides have different amino sugars.

What are some common examples of Aminoglycosides?

Common examples of aminoglycosides include streptomycin, neomycin, gentamicin, tobramycin, and amikacin. Being that their structure contains sugar moieties, these agents are very hydrophilic and hence are administered intravenously (IV).

What are some common examples of Tetracyclines?

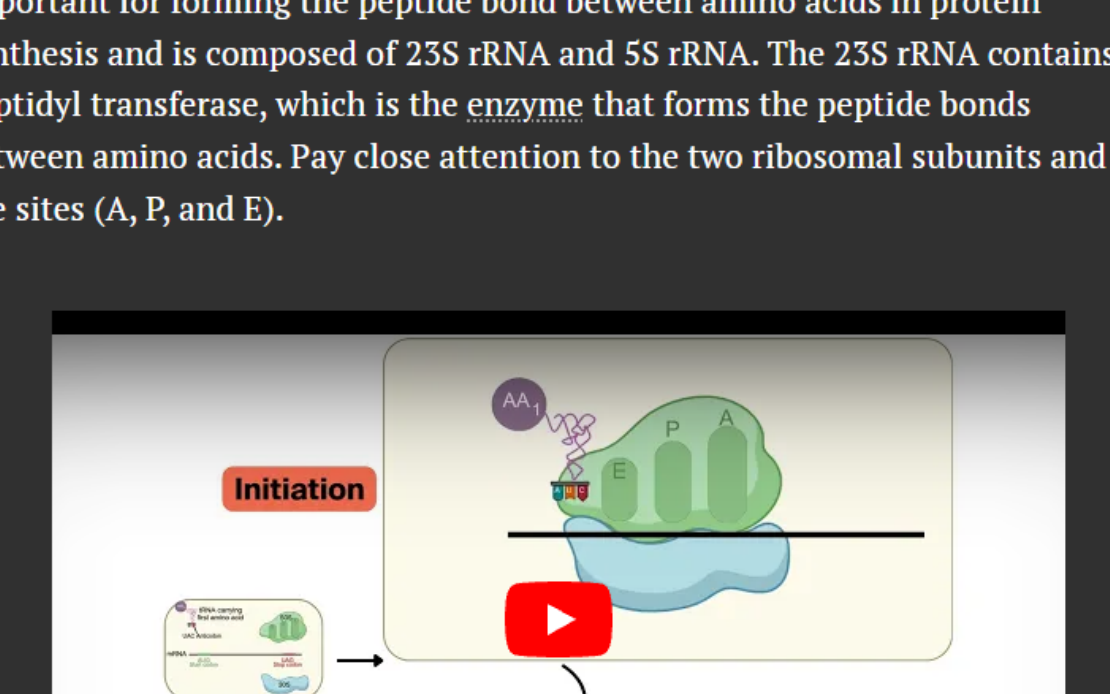
Tetracyclines (ending with '-cycline') were known for their broad-spectrum activity against bacteria. As bacteria developed resistance against them second-generation tetracyclines doxycycline and minocycline were developed with greater antibacterial properties and are still widely used today.

What are the types of 50S Protein Synthesis Inhibitors?

The 50S PSIs include Macrolides (suffix "-mycin", Azithromycin, Clarithromycin, Erythromycin), lincosamide (Clindamycin), streptogramins, and oxazolidinones (Linezolid) are important members of the 50S PSIs.

How Do Protein Synthesis Inhibitors Work?

To understand the mechanism of action of these agents it is important to understand the protein translation process in prokaryotes. The video (see below) walks you through the protein translation process. The 30S subunit itself is composed of 16S rRNA and 21 other proteins. The 50S subunit is important for forming the peptide bond between amino acids in protein synthesis and is composed of 23S rRNA and 5S rRNA. The 23S rRNA contains peptidyl transferase, which is the enzyme that forms the peptide bonds between amino acids. Pay close attention to the two ribosomal subunits and the sites (A, P, and E).



Is the inhibition selective for the bacteria?

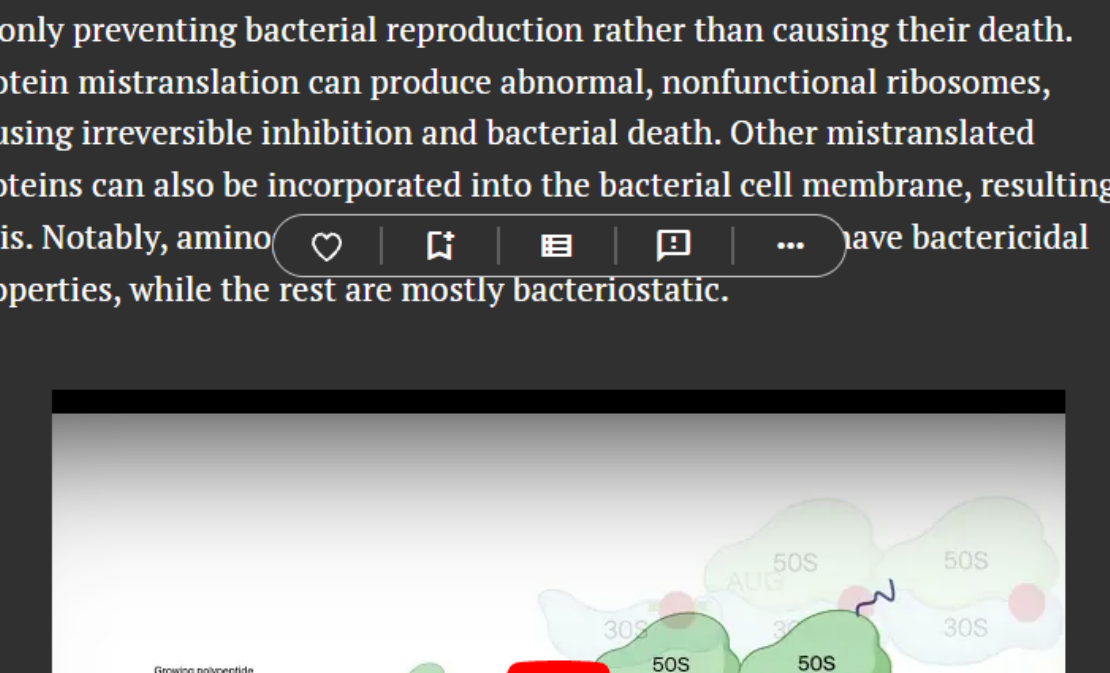
Note that eukaryotic ribosomes are not composed of 30S or 50S subunits, so PSI antibiotics do no[illegible]wever, this does not exclude 30S and 50S RIAs from having toxic side effects. For example, aminoglycosides are known for having a narrow therapeutic index, meaning that their therapeutic concentration range is close to or overlaps their toxic concentration range. Because of this, aminoglycosides are generally confined to inpatient use with careful monitoring.

What is the mechanism of action for 30S PSI?

Tetracyclines inhibit bacterial protein synthesis by inhibiting ribosomal 30S function. Tetracyclines bind to the 30S subunit and inhibit access to aminoacyl tRNA to the A (acceptor) site on the ribosomal complex. This binding is reversible (which makes tetracycline a bacteriostatic inhibitor), however, and is based on selective concentrations in susceptible bacteria.

The primary site of **Aminoglycosides** is the 30S ribosomal subunit and also targets the 16S rRNA ribosomal RNA. The mechanism of action is explained in the video below. The incorporation of the incorrect amino acids and proteins causes the cell membrane to lyse. Thus, aminoglycosides are categorized as bactericidal (NOTE: most PSIs are bacteriostatic).

Aminoglycosides disrupt translational proofreading of the amino acids, causing premature truncation or inaccurate translation of proteins. Aminoglycosides' specific MOA makes them bactericidal to certain bacterial species, meaning they kill them. Bactericidal is contrasted with bacteriostatic, which is defined as only preventing bacterial reproduction rather than causing their death. Protein mistranslation can produce abnormal, nonfunctional ribosomes, causing irreversible inhibition and bacterial death. Other mistranslated proteins can also be incorporated into the bacterial cell membrane, resulting in lysis. Notably, amino[illegible]have bactericidal properties, while the rest are mostly bacteriostatic.

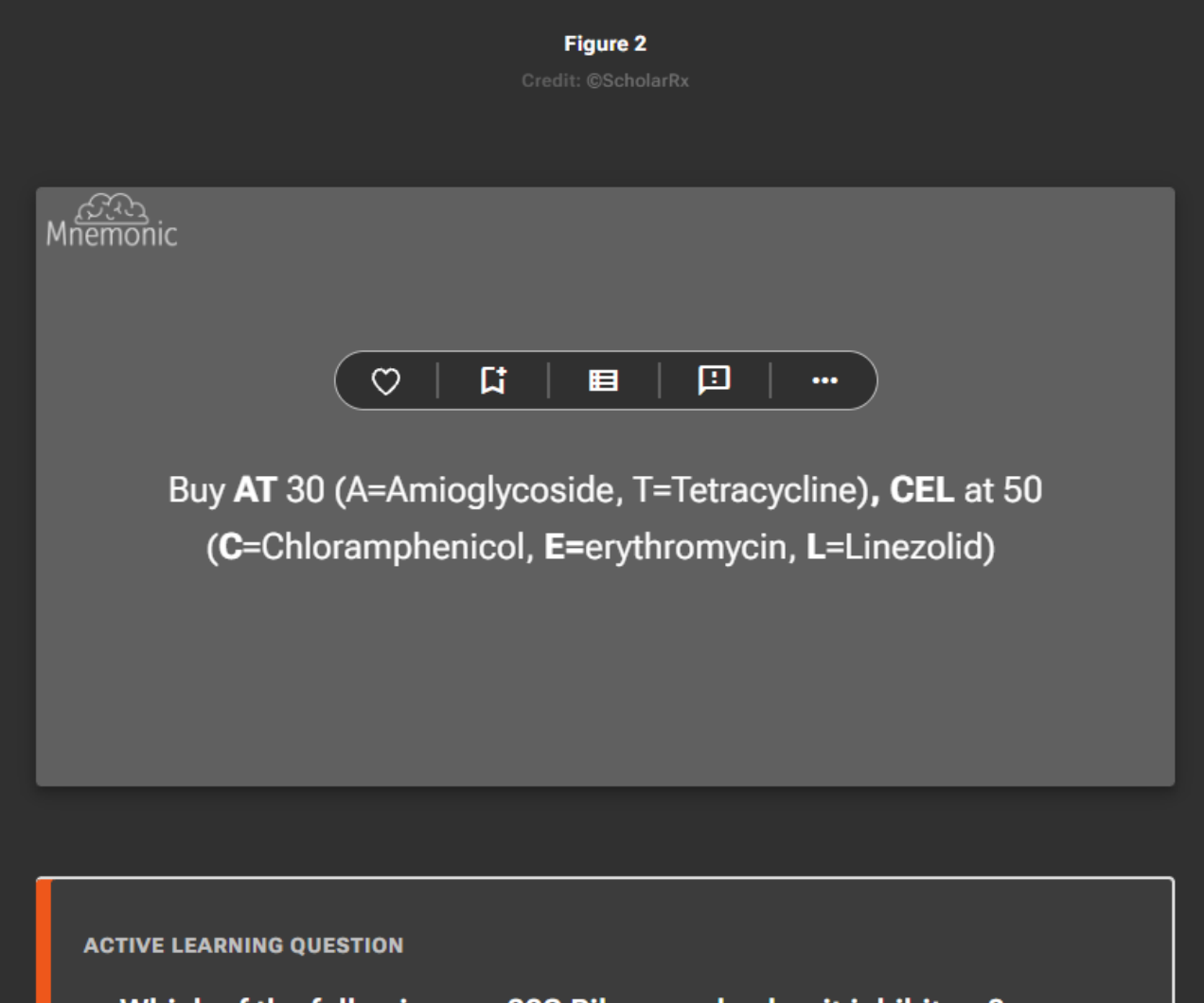


Are tetracyclines considered bacteriostatic or bactericidal?

What is the mechanism of action for 50S PSIs?

The mechanism of action of various 50S PSIs are described below: (Figure 2). The specific mechanism of action for the agents are as follows:

- **Macrolides and clindamycin** interfere with protein elongation by binding reversibly to the 50S ribosomal subunit. This binding inhibits the transpeptidation and translocation of tRNA across the A, P, and E sites of the ribosome, resulting in the release of incomplete proteins. Most of these agents are bacteriostatic in action.
- **Linezolid** binds to the interphase of the 30S and 50S subunits to prevent initiation complex formation by blocking the first peptide-bonding step. It binds to the 23S ribosomal RNA of the 50S subunit. This unique binding site acts as a deterrent for developing bacterial resistance. In fact, it allows linezolid to be free of cross-resistance that affects other protein synthesis inhibitors.
- The **streptogramins Dalfopristin** and **Quinupristin** work synergistically to block ribosomal functioning. Dalfopristin binds to the 50S and inhibits the early phase of bacterial protein synthesis. Quinupristin binds to the 50S ribosomal subunit to inhibit the late phase of protein synthesis. The synergistic action prevents proper functioning of the ribosomal subunits to disrupt the incorporation of amino acids into the peptide chains, leading to cell death.
- **Chloramphenicol** binds to the 50S ribosomal subunit and prevents transpeptidation, causing bacteriostatic action.



QUIZ

Figure 2

Credit: @[illegible]

Mnemonic

Buy **AT** 30 (**A**=Amioglycoside, **T**=Tetracycline), **CEL** at 50 (**C**=Chloramphenicol, **E**=erythromycin, **L**=Linezolid)

ACTIVE LEARNING QUESTION

Which of the following are 30S Ribosomal subunit inhibitors? (select all that apply)

- A. ☑ Gentamycin
- B. ☐ Clindamycin
- C. ☐ Linezolid
- D. ☐ Azithromycin
- E. ☑ Doxycycline

Explanation

The correct an[illegible]nd E). Gentamycin is an aminoglycoside, and Doxycycline is a Tetracycline class of antibiotics. They drugs bind to the 30S ribosomal subunit and inhibit protein synthesis. Clindamycin (B) is a lincosamide that binds to the 50S ribosomal subunit to inhibit protein synthesis. Linezolid (C) binds to the 50S ribosomal subunit to inhibit protein transpeptidation. Azithromycin (D) is a macrolide that binds to the 50S ribosomal subunit to inhibit protein synthesis.

How Do Bacteria Resist Protein Synthesis Inhibitors?

Bacterial resistance can be intrinsic, acquired, or transmissible. Intrinsic resistance is pre-existing bacterial resistance prior to antibiotic exposure, while acquired describes resistance that is gained after exposure. Transmissible resistance is transferrable between bacteria, often by horizontal gene transfer of plasmids. Common acquired and transmissible mechanisms of resistance include efflux pumps, enzymatic inactivation, and target site alteration.

Enzymatic Inactivation

Bacteria are capable of changing the antibiotic itself. They can produce enzymes (e.g., phosphotransferase or adenylyltransferase) that inactivate PSIs by modifying their chemical structure. Gram-negative organisms are responsible for health care–associated infections have been well documented for spreading enzymes that inactivate aminoglycosides by acetylation to prevent their interaction with the 16s rRNA.

Target Site Alteration

Bacteria can decrease the affinity of their own ribosomes for PSIs. For example, methylation to parts of the 50S subunit can reduce the binding capability of certain 50S RIAs. Specifically, gram-positive bacteria can transfer *erm* (erythromycin ribosome methylation) genes, which alter the 23S rRNA. Changing this target site creates resistance not only to macrolides, but also to the rest of the MLS family (including clindamycin, a lincosamide) due to their shared target site.

Efflux Pumps

Bacteria can actively transport protein synthesis inhibitors out of their cytoplasm to reduce antibiotic activity. Remember that PSIs need to act on the ribosome within the cytoplasm to be effective. Tetracyclines have been known to be susceptible to efflux pumps due to the TetA efflux pump protein. This antiport pump inserts into the cell membrane to pump tetracycline out and protons inside the cell. Tetracyclines, however, remain effective against some zoonotic species. Efflux pumps can target a specific drug or multiple drugs and are commonly found in many bacteria today.

Each PSI has a different mechanism of action with an accompanying different method of resistance. Taking this idea one step further, bacteria can be resistant to some PSIs but not others. Due to differing resistance to PSIs, each drug has a unique spectrum of activity. In the next brick, we are going to discuss the spectrum of activity for each of these antibiotics.

What pharmacokinetic parameters are important for PSIs? (LO#2)

Pharmacokinetics (PK) plays an important role in effective use of PSIs. Let's first review the 30S PK profiles. The 30S RIAs are not substantially metabolized. Due to poor oral absorption, aminoglycosides are administered IV or intramuscularly (IM) and are eliminated unchanged by the kidneys. Tetracyclines can be taken orally, IV, or IM and are eliminated via biliary excretion in urine and feces.

How are aminoglycosides administered and eliminated?

The 50S RIAs can generally be administered orally or topically and are hepatically metabolized. We will discuss the relevance of the hepatic metabolism in the context of drug interactions in one of the following bricks (link to PSI-3).

The 30S RIAs have generally good bioavailability but minimal CSF penetration. Aminoglycosides hav[illegible]rotein binding and water solubility. Tetracyclines have great tissue penetration and distribution due to their lipophilicity.

The 50S RIAs also have good bioavailability with variable CSF penetration. Orally administered macrolides have a bioavailability of 30% to 50% depending on the stability of the particular drug at gastric pH. All macrolides have good penetration, making them well concentrated in tissues and phagocytic cells but unable to penetrate CSF well in the absence of meningeal inflammation. Clindamycin has approximately 90% bioavailability after oral administration and distributes well into body tissues, especially bone. It also has little CSF penetration. Linezolid has excellent bioavailability and readily enters the CSF.

CLINICAL CORRELATION

As mentioned earlier, aminoglycosides are only administered IV. They also have a narrow therapeutic window that limits them to inpatient use. Aminoglycoside dosing can be separated into traditional dosing or extended-interval dosing with special considerations for synergistic therapy. Extended-interval dosing involves taking one large dose daily. Traditional dosing uses multiple small doses of a drug throughout the day. Therapeutic monitoring of aminoglycoside toxicity is measured every 1-3 days via serum creatinine and subjective patient assessment of vestibular and auditory function. When used synergistically with other antibiotic agents or in patients with reduced renal function (e.g., elderly patients and patients with renal disease or who are taking concomitant nephrotoxic drugs), lower serum concentrations are targeted.

BACK TO INTRODUCTION ↑

CASE CONNECTION

We will discuss the answer at the end of the Protein Synthesis Inhibitors-2 brick.

Summary

- Protein synthesis inhibitor (PSI) antibiotics interact with the Acceptor, Peptidyl, and Elongation sites or bind to other parts of the ribosomal subunits to inhibit bacterial growth or kill the bacteria.
- Aminoglycosides and tetracyclines are 30S PSIs, while Macrolide, lincosamide, streptogramins, and oxazolidinones are 50S PSIs.
- PSIs are most effective during the acute stage of an infection when bacteria are actively dividing.
- Combination therapy with antibiotics can enhance effectiveness, prevent emergence of resistant organisms, or broaden the antibacterial spectrum.
- Bacterial resistance to PSIs can be intrinsic, acquired, or transmissible, with common mechanisms including efflux pumps, enzymatic inactivation, and target site alteration.
- Pharmacokinetics, including administration method, metabolism, and elimination, plays a crucial role in the effective use of PSIs.

Review Questions

1. Which antibiotic binds to the 23S subunit of the 50S ribosomal unit to inhibit protein synthesis?

- A. ○ Azithrom[illegible]
- B. ○ Gentamycin
- C. ● Linezolid
- D. ○ Minocycline

Hide Explanation

The correct answer is Linezolid (C). Linezolid binds to the 23S site on the 50S bacterial ribosomal subunit to inhibit protein synthesis. While Azithromycin (A) binds to the 50S subunit, it has not been shown to target the 23S binding site. Gentamycin (an Aminoglycoside) (B) and Minocycline (D) are 30S binding protein synthesis inhibitors.

2. Which antibiotic is cleared unchanged renaly?

- A. ○ Azithromycin
- B. ○ Clarithromycin
- C. ○ Minocycline
- D. ● Gentamycin

Hide Explanation

The correct answer is Gentamycin (D). Gentamycin, an aminoglycoside is hydrophilic and is cleared entirely with renal excretion. Azithromycin (A) and Clarithromycin (B) are cleared hepatically. Minocycline (D) is cleared in the feces.

3. Which antibiotic is a 50S ribosomal subunit inhibitor?

- A. ○ Amikacin
- B. ● Dalfopristin/Quinupristin
- C. ○ Doxycycline
- D. ○ Gentamycin

Hide Explanation

The correct answer is Dalfopristin (B). Dalfopristin works synergistically with Quinupristin to inhibit protein synthesis by binding to the bacterial 50S ribosomal subunit. Amikacin (A) and Gentamycin (D) are aminoglycosides that bind to 30S subunit to inhibit protein synthesis. Doxycycline (C) binds to the 30S subunit to inhibit protein synthesis.

How would you rate this Brick?

subunit. Amikacin (A) and Gentamycin (D) are aminoglycosides that bind to 30S subunit to inhibit protein synthesis. Doxycycline (C) binds to the 30S subunit to inhibit protein synthesis.

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action and mechanism of resistance of protein synthesis inhibitors.~~
02 ☑ ~~Discuss key pharmacokinetic parameters that are considered when using the protein synthesis inhibitors.~~
03 ☑ ~~Recognize the common adverse reactions associated with protein synthesis inhibitors.~~
04 ☑ ~~Identify the common drug interactions associated with protein synthesis inhibitors.~~

Objective Confidence: 4/4